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Ingot growth apparatus

Abstract

An ingot growth apparatus is disclosed. The ingot growth apparatus according to an embodiment of the present invention may comprise: a growth furnace having a main crucible which is disposed inside the growth furnace and in which molten silicon is held in order to grow an ingot; a susceptor formed to surround the outer surface of the main crucible and heating the main crucible; a heater formed to surround the outer surface of the susceptor and including a coil which is supplied with power to generate a magnetic field and heats the susceptor by electromagnetic induction from the magnetic field; and a heat insulation member disposed between the coil and the susceptor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2020-0126319, filed on Sep. 28, 2020, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present invention relates to an apparatus for growing an ingot.

BACKGROUND ART

(3) Single crystal silicon is used as a basic material for most solar light and semiconductor components, and these materials are manufactured as single crystals with high purity, and one of the manufacturing methods thereof is the Czochralski method.

(4) In the Czochralski crystallization method, silicon is placed into a crucible, and the crucible is heated to melt the silicon. In addition, when a single crystal seed is pulled upward while rotating in a state of being in contact with the molten silicon, an ingot having a predetermined diameter is grown.

(5) The continuous Czochralski method (CCz), which is one of the Czochralski methods, is a method of continuously growing an ingot while supplementing the consumed molten silicon by continuously injecting solid polysilicon or molten silicon into the crucible.

(6) In order to grow an ingot through the continuous Czochralski method (CCz), it is important to secure the efficiency of power energy consumed to heat a crucible.

DISCLOSURE

Technical Problem

(7) According to an exemplary embodiment of the present invention, it is directed to providing an apparatus for growing an ingot that improves the efficiency of power energy for heating a crucible.

Technical Solution

(8) The apparatus for growing an ingot according to an exemplary embodiment of the present invention may include a growth furnace having a main crucible which is disposed inside the growth furnace and in which molten silicon is held in order to grow an ingot; a susceptor which is formed to surround the outer surface of the main crucible and heats the main crucible; a heater which is formed to surround the outer surface of the susceptor and includes a coil which is supplied with power to generate a magnetic field and heats the susceptor by electromagnetic induction from the magnetic field; and a heat insulation member which is disposed between the coil and the susceptor.

(9) In this case, the heat insulation member may be made of a plate-shaped member which is formed on the outer side surface of the susceptor, and wherein one side surface of the heat insulating member in the transverse direction and the other side surface of the heat insulating member in the transverse direction may be disposed to face each other and be spaced apart from one side of the outer side surface of the susceptor.

(10) In this case, a non-magnetic material may be provided between one side surface of the heat insulating member in the transverse direction and the other side surface of the heat insulation member in the transverse direction.

- (11) In this case, the heater further may include a shield which is formed to surround the outer side surface of the coil and blocks exposure of the coil to the inner space of the growth furnace.
- (12) In this case, the heat insulation member may be disposed on the upper side surface of the shield and may be formed to partially cover the upper side surface of the shield.
- (13) In this case, the shield may be made of ceramic.
- (14) In this case, the heat insulation member may be made of carbon fiber.
- (15) In this case, the apparatus for growing an ingot may further include a cover member which blocks exposure of the heat insulation member to the inner space of the growth furnace.
- (16) In this case, the cover member may be made of a ceramic material.
- (17) In this case, the cover member may include at least one of alumina (Al.sub.2O.sub.3), zirconia (ZrO.sub.2), silica (SiO.sub.2) or silicon nitride (Si.sub.3N.sub.4).
- (18) In this case, the coil may include a cooling pipe through which cooling water flows inside the coil.
- (19) In this case, the thickness of the heat insulation member may correspond to twice the thickness of the susceptor.

Advantageous Effects

- (20) In the apparatus for growing an ingot according to an exemplary embodiment of the present invention, since the heat insulation member blocks the heat of a susceptor, which has been heated, from being transferred to a coil, it is possible to prevent the coil from being damaged by the heat.
- (21) In addition, since the heat insulation member blocks the heat of a susceptor from being transferred to cooling water flowing inside the coil, it is possible to increase the efficiency of power energy for heating the susceptor.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a view schematically showing the apparatus for growing an ingot according to an exemplary embodiment of the present invention.
- (2) FIG. 2a is a cross-sectional view mainly showing the heater, the heat insulation member and the cover member of FIG. 1.
- (3) FIG. 2b is a view showing a state in which the heat insulation member of FIG. 1 is viewed from above.
- (4) FIG. 3 is a diagram showing the temperature of a hotzone of the apparatus for growing an ingot according to an exemplary embodiment of the present invention through simulation.
- (5) FIG. 4 is a diagram showing the temperature of a hotzone of the apparatus for growing an ingot through simulation, when there is no heat insulation member.
- (6) FIG. 5 is a graph showing the temperature of a susceptor according to the material of a shield and the presence of a heat insulation member through simulation.
- (7) FIGS. 6 to 8 are diagrams showing the susceptor temperature and the ambient temperature of a susceptor according to the distance between the coil and the susceptor.
- (8) FIGS. 9 to 11 are diagrams showing the susceptor temperature and the ambient temperature of a susceptor according to the thickness of an insulation member through simulation.
- (9) FIG. 12 is a diagram showing the susceptor temperature and the ambient temperature of a susceptor through simulation, when a shield heat insulation member is disposed below the coil.
- (10) FIG. 13 is a diagram showing the susceptor temperature and the ambient temperature of a susceptor through simulation, when a shield heat insulation member is not disposed below the coil.
- (11) TABLE-US-00001 100: Apparatus for growing ingot 110: Growth furnace 120: Main crucible 130: Susceptor 140: Heater 141: Coil 143: Heat insulation member

MODES OF THE INVENTION

- (12) Terms and words used in the present specification and claims should not be construed as limited to their usual or dictionary definition, and they should be interpreted as a meaning and concept consistent with the technical idea of the present invention based on the principle that inventors may appropriately define the terms and concept in order to describe their own invention in the best way.
- (13) Accordingly, the exemplary embodiments described in the present specification and the configurations shown in the drawings correspond to preferred exemplary embodiments of the present invention, and do not represent all the technical spirit of the present invention, and thus, the configurations may have various examples of equivalent and modification that can replace them at the time of filing the present invention.
- (14) It is understood that the terms “include” or “have”, when used in the present specification, are intended to describe the presence of stated features, integers, steps, operations, elements, components and/or a combination thereof but do not preclude the possibility of the presence or addition of one or more other features, integers, steps, operations, elements, components or a combination thereof.
- (15) The presence of an element in/on “front”, “rear”, “upper or above or top” or “lower or below or bottom” of another element includes not only being disposed in/on “front”, “rear”, “upper or above or top” or “lower or below or bottom” directly in contact with other elements, but also cases in which another element being disposed in the middle, unless otherwise specified. In addition, unless otherwise specified, that an element is “connected” to another element includes not only direct connection to each other but also indirect connection to each other.
- (16) Hereinafter, the apparatus for growing an ingot according to an exemplary embodiment of the present invention will be described with reference to the drawings. In the present specification, in terms of describing the apparatus for growing an ingot according to an exemplary embodiment of the present invention, the configurations that are not related to the contents of the present invention are not illustrated in detail or omitted for the sake of simplification of the drawings, and the apparatus for growing an ingot according to the present invention will be described by mainly focusing on the contents that are related to the spirit of the invention.
- (17) In the present specification, the arrow direction of the Z-axis is referred to as an upward direction of the growth furnace. The downward direction means a direction opposite to the upward direction.
- (18) FIG. 1 is a view schematically showing the apparatus for growing an ingot according to an exemplary embodiment of the present invention.
- (19) Referring to FIG. 1, the apparatus for growing an ingot **100** according to an exemplary embodiment of the present invention may include a growth furnace **110**, a main crucible **120**, a susceptor **130**, a heater **140** and a heat insulation member **143**.
- (20) The growth furnace **110** has an internal space **110a** which is maintained in a vacuum state, and is formed such that an ingot I is grown in the internal space **110a**. A main crucible **120** to be described below is disposed in the inner space **110a**.
- (21) The growth furnace **110** is provided with a vacuum pump (not illustrated) and an inert gas supply part (not illustrated). The vacuum pump may maintain the internal space **110a** in a vacuum atmosphere. In addition, the inert gas supply part supplies inert gas to the internal space **110a**. The inert gas may be, for example, argon (Ar).
- (22) The main crucible **120** is accommodated in the inner space **110a** of the growth furnace **110**. The main crucible **120** may accommodate molten silicon M. The main crucible **120** is generally formed in the shape of a reverse dome. In addition, the main crucible **120** is not limited to being formed in the shape of a reverse dome, and may be formed in various shapes such as a cylindrical shape.
- (23) In addition, the main crucible **120** is made of a quartz material. However, the main crucible **120** is not limited to being made of a quartz material, and may include various materials that have

heat resistance at a temperature of about 1,400° C. or higher and withstand a sudden change in temperature.

(24) In addition, while a single crystal seed S is in contact with the molten silicon M accommodated in the main crucible **120**, when a wire W which is connected to the upper side of the growth furnace **110** pulls up the single crystal seed S in the upward direction (Z axis), an ingot I having a predetermined diameter is grown along the pulling direction (Z axis) of the ingot I.

(25) In addition, the growth furnace **110** is provided with a preliminary melting part (not illustrated) for receiving and melting a solid silicon raw material. The preliminary melting part supplies the molten silicon to the main crucible **120**.

(26) The susceptor **130** surrounds the outer surface of the main crucible **120**. The susceptor **130** supports the main crucible **120**. The inner surface of the susceptor **130** has a shape corresponding to the outer surface of the main crucible **120**. For example, if the main crucible **120** has a reverse dome shape, the susceptor **130** also has a reverse dome shape. The susceptor **130** is made of a graphite material. In addition, the susceptor **130** is not limited to being made of a graphite material, and may include various materials having strong heat resistance and conductor properties.

(27) Accordingly, even if the main crucible **120** is made of a quartz material and deformed at a high temperature, the susceptor **130** surrounds and supports the main crucible **120** so as to maintain a state in which the main crucible **120** receives the molten silicon M.

(28) In addition, a susceptor support part **150** for supporting the susceptor **130** is disposed on the lower side **112** of the growth furnace **110**. The upper end of the susceptor support **150** has a shape corresponding to the lower end of the susceptor **130**. In addition, while the susceptor support part **150** supports the susceptor **130** at the lower side of the growth furnace **110**, the susceptor support part **150** rotates together with the susceptor **130**. Accordingly, while the main crucible **120** accommodates the molten silicon M, the main crucible **120** is rotated together with the susceptor **130**.

(29) In addition, the growth furnace **110** is provided with a driving part (not illustrated) that provides a rotational force to rotate the susceptor support **150**. The susceptor support part **150** is rotatably connected to the driving part. When the driving part receives power and provides a rotational force to the susceptor support part **150**, the main crucible **120** is rotated together with the susceptor **130**.

(30) In addition, a heater **140** for heating the susceptor **130** is provided in the growth furnace **110**. The heater **140** includes a coil **141** for receiving power to generate a magnetic field and a shield **142** for surrounding the coil **141**.

(31) The coil **141** is formed to surround the outer surface of the susceptor **130**. The coil **141** receives power to generate a magnetic field. In addition, the coil **141** generates a current in the susceptor **130** by electromagnetic induction by a magnetic field. In this case, the current generated in the susceptor **130** is converted into thermal energy. Accordingly, the heater **140** heats the susceptor **130**. The heat of the susceptor **130** is conducted to the main crucible **120**, and the susceptor **130** heats the main crucible **120**.

(32) The shield **142** supports the coil **141** such that the coil **141** is maintained in a certain shape. The shield **142** is made of ceramic. For example, the shield **142** may include at least one of alumina (Al.sub.2O.sub.3), zirconia (ZrO.sub.2), silica (SiO.sub.2) or silicon nitride (Si.sub.3N.sub.4).

(33) The shield **142** blocks the coil from being exposed to the inner space **110a** of the growth furnace **110**. Accordingly, the shield **142** blocks the coil **141** from being exposed to the inner space **110a** of the growth furnace **110** such that when the coil **141** receives power to form a magnetic field, it is possible to prevent an arc discharge from occurring due to plasma phenomenon in the vacuum state or an arc discharge from being generated by the coil **141** coming into contact with inert gas (e.g., argon) which is present in the inner space **110a**.

(34) In addition, a heater support part **160** supporting the heater **140** is disposed below the growth furnace **110**. The heater support part **160** is generally formed in a cylindrical shape. The susceptor

support **150** is disposed inside the heater support part **160** having the cylindrical shape. In addition, the upper end of the heater support part **160** has a shape corresponding to the lower end of the heater **140**, and the heater **140** is disposed on the upper end of the heater support part **160**.

(35) A heat insulation member **143** is provided between the susceptor **130** and the coil **141** to block the movement of heat. In addition, the heat insulation member **143** is surrounded by the cover member **144** to prevent exposure to the inner space **110a** of the growth furnace **110**.

(36) The heat insulation member **143** and the cover member will be described in detail below with reference to the drawings.

(37) FIG. **2a** is a cross-sectional view mainly showing the heater, the heat insulation member and the cover member of FIG. **1**, and FIG. **2b** is a view showing a state in which the heat insulation member of FIG. **1** is viewed from above.

(38) Referring to FIGS. **2a** and **2b**, the heat insulation member **143** is disposed on the upper surface of the shield **142** and is formed to partially cover the upper surface of the shield **142**. In addition, the heat insulation member **143** is formed to partially surround the outer surface of the susceptor **130**. In this case, the heat insulation member **143** may be formed of a plate-shaped member which is disposed between the shield **142** and the susceptor **130**.

(39) Meanwhile, a cooling pipe **145** through which cooling water flows is formed inside the coil **141**. The temperature of the cooling water of the cooling pipe **145** is approximately 300K. The cooling water flowing along the cooling pipe **145** cools the heat generated from the coil **141**.

(40) The heat insulation member **143** blocks the heat of the susceptor **130** from being transferred to the coil **141**. In addition, the heat insulation member **143** blocks the heat of the susceptor **130** from being transferred to cooling water flowing inside the coil **141**. Accordingly, the heat of the susceptor **130** is transferred to the main crucible **120**, thereby increasing the efficiency of power energy for heating the main crucible **120**.

(41) As illustrated in FIG. **2b**, one side surface **143a** of the heat insulation member **143** in the traverse direction and the other side surface **143b** of the heat insulation member **143** in the traverse direction are disposed to face each other and be spaced apart from the outer side surface of the susceptor **130**. In addition, at least one cut-out area is formed in the heat insulation member **143**. That is, the heat insulation member **143** is not formed in a loop shape. Accordingly, one side **143a** of the heat insulation member **143** and the other side **143b** of the heat insulation member **143** are not electrically connected, and thus, the heat insulation member **143** is prevented from being heated by electromagnetic induction.

(42) In addition, according to various exemplary embodiments of the present invention, the heat insulation member **143** may be provided as a plurality of heat insulation members that are separated from each other by a plurality of cut areas. The plurality of heat insulation members are electrically insulated from each other by the plurality of cut-out areas.

(43) In addition, the heat insulation member **143** is made of carbon fiber. In addition, the heat insulation member **143** is formed in the form of a felt including carbon fibers. In addition, according to various exemplary embodiments of the present invention, the heat insulation member **143** may have a structure in which a felt including carbon fibers is laminated. In addition, according to various exemplary embodiments of the present invention, the heat insulation member **143** may be formed of zirconia (ZrO.sub.2) or silica (SiO.sub.2) in powder form.

(44) The cover member **144** is formed to surround the heat insulation member **143**. In addition, the cover member **144** is formed to cover a portion of the shield **142** adjacent to the heat insulation member **143**. In addition, the cover member **144** fills a space between the susceptor **130** and the heat insulation member **143**. Accordingly, since the cover member **144** blocks the heat insulation member **143** from being exposed to the inner space **110a** (refer to FIG. **1**) of the growth furnace **110** (refer to FIG. **1**), it is possible to prevent an arc discharge from occurring due to plasma phenomenon in a vacuum state or prevent an arch discharge from being generated by the heat insulation member **143** coming into contact with inert gas which is present in the inner space **110a**.

(45) In addition, the cover member **144** is made of ceramic. For example, the cover member **144** is made of any one of alumina (Al.sub.2O.sub.3), zirconia (ZrO.sub.2), silica (SiO.sub.2) or silicon nitride (Si.sub.3N.sub.4). In addition, according to various exemplary embodiments of the present invention, the cover member **144** may be made of the same material as the shield **142**.

(46) In addition, according to various exemplary embodiments of the present invention, the cover member **144** may be formed of a coating layer to cover a portion of the shield **142** adjacent to the heat insulation member **143** while covering the heat insulation member **143**.

(47) A non-magnetic material **146** is provided between the one side surface **143a** of the heat insulation member **143** in the transverse direction and the other side surface **143b** of the heat insulation member **143** in the transverse direction. The non-magnetic material **146** electrically insulates between one side **143a** of the heat insulation member **143** and the other side **143b** of the heat insulation member **143**.

(48) In addition, the non-magnetic material **146** is made of ceramic. For example, the non-magnetic material **146** is made of any one of alumina (Al.sub.2O.sub.3), zirconia (ZrO.sub.2), silica (SiO.sub.2) or silicon nitride (Si.sub.3N.sub.4).

(49) In addition, the distance *e* between one side **143a** of the heat insulation member **143** and the other side **143b** of the heat insulation member **143** is approximately 0.5 mm or more. For example, when the distance *e* is approximately 0.5 mm, the thickness of the non-magnetic material **146** is also 0.5 mm.

(50) In addition, according to various exemplary embodiments of the present invention, a plurality of cut areas may be formed in the heat insulation member **143**, and the non-magnetic material **146** may be filled in the plurality of cut areas.

(51) In an exemplary embodiment of the present invention, the thickness *t* of the heat insulation member is 40 to 60 mm. The thickness *t* of the heat insulation member will be described in detail later below reference to the drawings.

(52) FIG. 3 is a diagram showing the temperature of a hotzone of the apparatus for growing an ingot according to an exemplary embodiment of the present invention through simulation, and FIG. 4 is a diagram showing the temperature of a hotzone of the apparatus for growing an ingot through simulation, when there is no heat insulation member.

(53) A hotzone of the apparatus for growing an ingot **100** (refer to FIG. 1) is defined as an area including the susceptor **130** (refer to FIG. 1) and the surrounding components of the susceptor **130** (refer to FIG. 1).

(54) Referring to FIGS. 3 and 4, the heating performance of the susceptor **130** according to the presence of the heat insulation member **143** (refer to FIG. 2a) will be examined.

(55) When FIG. 3 and FIG. 4 are compared, if the heat insulation member is disposed between the coil and the susceptor, the temperature of the side portion of the susceptor is close to about 1,695° C., but if the heat insulation member is not disposed between the coil and the susceptor, the temperature of the susceptor does not exceed approximately 1,385° C. Since the melting point of silicon corresponds to about 1,414° C., when the heat insulation member is not disposed, more power energy is consumed to increase the temperature of the susceptor.

(56) As such, in the apparatus for growing an ingot according to an exemplary embodiment of the present invention, as the heat insulation member is disposed between the coil and the susceptor, the efficiency of power energy for increasing the temperature of the susceptor is increased.

(57) FIG. 5 is a graph showing the temperature of a susceptor according to the material of a shield and the presence of a heat insulation member through simulation.

(58) Referring to FIG. 5, cases where the material of the shield (e.g., castable) is zirconia (ZrO.sub.2) and cases where the material of the shield (e.g., castable) is alumina (Al.sub.2O.sub.3) will be compared to examine the effects thereof on the temperature of the susceptor.

(59) In all cases of where the material of the shield (e.g., castable) is zirconia (ZrO.sub.2) and cases where the material of the shield (e.g., castable) is alumina (Al.sub.2O.sub.3), the temperature

distributions according to the distance (Susceptor Upper Distance) from the upper side of the susceptor are similar.

(60) Meanwhile, as shown in FIGS. 3 and 4, when the heat insulation member is not disposed (e.g., ZrO.sub.2 Removed Felt, Al.sub.2O.sub.3Removed Felt), the temperature of the susceptor is lower than the temperature of the susceptor when the heat insulation member is disposed regardless of the material of the shield.

(61) As such, the heat insulation member of the apparatus for growing an ingot according to an exemplary embodiment of the present invention increases the heat generating performance of the susceptor, thereby reducing the consumption of power energy for heating the susceptor.

(62) FIGS. 6 to 8 are diagrams showing the susceptor temperature and the ambient temperature of a susceptor according to the distance between the coil and the susceptor.

(63) The coil illustrated in FIG. 6 is disposed to be spaced apart from the susceptor by a first distance, the coil illustrated in FIG. 7 is disposed to be spaced apart from the susceptor by a second distance, and the coil illustrated in FIG. 8 is disposed to be spaced apart from the susceptor by a third distance. The second distance is longer than the first distance, and the third distance is longer than the second distance.

(64) Heat insulation members having the same thickness are disposed between the coils illustrated in FIGS. 6 to 8 and the susceptor.

(65) When FIGS. 6 to 8 are compared, the susceptor illustrated in FIG. 6 generates heat at a higher overall temperature than the susceptors illustrated in FIGS. 7 and 8. This is because, as the distance between the coil and the susceptor increases, the heat of the susceptor is transferred to a space between the coil and the susceptor, and the heat loss of the susceptor increases.

(66) In addition, the power energy for heating the susceptor illustrated in FIG. 6 is 49.75 kW, the power energy for heating the susceptor illustrated in FIG. 7 is 53.47 kW, and the power energy for heating the susceptor illustrated in FIG. 8. is 51.04 kW. As such, when the heat insulation member is disposed between the coil and the susceptor, as the coil approaches the susceptor, power energy for heating the susceptor is reduced.

(67) FIGS. 9 to 11 are diagrams showing the susceptor temperature and the ambient temperature of a susceptor according to the thickness of an insulation member through simulation.

(68) The coil illustrated in FIG. 9 is disposed to be spaced apart from the susceptor by a fourth distance, the coil illustrated in FIG. 10 is disposed to be spaced apart from the susceptor by a fifth distance, and the coil illustrated in FIG. 11 is disposed to be spaced apart from the susceptor by a sixth distance. The fifth distance is longer than the fourth distance, and the sixth distance is longer than the fifth distance.

(69) The thickness of the heat insulation member illustrated in FIG. 9 is 10 mm, the thickness of the heat insulation member illustrated in FIG. 10 is 30 mm, and the thickness of the heat insulation member illustrated in FIG. 11 is 50 mm.

(70) In addition, the power energy for heating the susceptor illustrated in FIG. 9 is 65.76 kW, the power energy for heating the susceptor illustrated in FIG. 10 is 53.88 kW, and the power energy for heating the susceptor illustrated in FIG. 11. is 45.95 kW.

(71) As shown in FIGS. 6 to 8, as the coil moves away from the susceptor, power energy for heating the susceptor increases, but as the thickness of the insulation member increases, power energy for heating the susceptor is reduced. That is, it can be seen that, in order to reduce the power energy for heating the susceptor, increasing the thickness of the heat insulation member is an important design consideration than increasing the distance between the coil and the susceptor.

(72) Meanwhile, according to an exemplary embodiment of the present invention, the thickness of the heat insulation member may correspond to approximately twice the thickness of the susceptor. For example, when the thickness of the susceptor is 25 mm, the thickness of the heat insulation member is 50 mm. When the thickness of the heat insulation member is less than twice the thickness of the susceptor, the consumption of power energy is increased. Conversely, when the

thickness of the heat insulation member is greater than twice the thickness of the susceptor, the consumption of power energy is not increased, but the cost of the heat insulation member is increased.

(73) FIG. 12 is a diagram showing the susceptor temperature and the ambient temperature of a susceptor through simulation, when a shield heat insulation member is disposed below the coil, and FIG. 13 is a diagram showing the susceptor temperature and the ambient temperature of a susceptor through simulation, when a shield heat insulation member is not disposed below the coil.

(74) According to another exemplary embodiment of the present invention, as illustrated in FIG. 12, a shield heat insulation member 148 is provided on the lower side of the coil to block heat transfer.

(75) The temperature of the susceptor and the ambient temperature of the susceptor when the shield heat insulation member 148 illustrated in FIG. 12 is disposed are higher than the temperature of the susceptor and the ambient temperature of the susceptor when the shield heat insulation member illustrated in FIG. 13 is not disposed.

(76) In addition, the power energy for heating the susceptor illustrated in FIG. 12 is 45.98 kW, and the power energy for heating the susceptor illustrated in FIG. 13 is 49.75 kW. This is because the shield heat insulation member 148 blocks heat radiated from the susceptor. That is, the power energy for heating the susceptor when the shield heat insulation member 148 is disposed has the effect of saving approximately 7.6% compared to the power energy for heating the susceptor when the shield heat insulation member is not disposed.

(77) Although the exemplary embodiments of the present invention have been described, the spirit of the present invention is not limited by the exemplary embodiments presented herein, and a person skilled in the art who understands the spirit of the present invention may easily suggest other exemplary embodiments by modifying, changing, deleting or adding components within the scope of the same spirit, but this will also be within the spirit of the present invention.

Claims

1. An apparatus for growing an ingot, comprising: a growth furnace having a main crucible which is disposed inside the growth furnace and in which molten silicon is held in order to grow the ingot; a susceptor which is formed to surround an outer surface of the main crucible and heats the main crucible; a heater which is formed to surround the outer surface of the susceptor and includes a coil which is supplied with power to generate a magnetic field and heats the susceptor by electromagnetic induction from the magnetic field; and a heat insulation member which is disposed between the coil and the susceptor; wherein the heat insulation member includes at least one cut-out area formed such that one side surface of the heat insulating member in a transverse direction, which is a circumferential direction of the heat insulation member when viewed from above, and the other side surface of the heat insulating member in the transverse direction of the heat insulation member are spaced apart from each other in the transverse direction; and wherein a non-magnetic material is positioned between the one side surface of the heat insulating member and the other side surface of the heat insulation member to electrically insulate the two side surfaces from each other.

2. The apparatus of claim 1, wherein the heat insulation member is made of a plate-shaped member which is formed on an outer side surface of the susceptor, and wherein one side surface of the heat insulating member in the transverse direction and the other side surface of the heat insulating member in the transverse direction are disposed to one side of the outer side surface of the susceptor.

3. The apparatus of claim 1, wherein the heater further comprises a shield which is formed to surround an outer side surface of the coil and blocks exposure of the coil to an inside of the growth furnace.

4. The apparatus of claim 3, wherein the heat insulation member is disposed on an upper side surface of the shield and is formed to partially cover the upper side surface of the shield.
 5. The apparatus of claim 3, wherein the shield is made of ceramic.
 6. The apparatus of claim 1, wherein the heat insulation member is made of carbon fiber.
 7. The apparatus of claim 1, further comprising: a cover member which blocks exposure of the heat insulation member to an inside of the growth furnace.
 8. The apparatus of claim 7, wherein the cover member is made of a ceramic material.
 9. The apparatus of claim 7, wherein the cover member comprises at least one of alumina (Al.sub.2O.sub.3), zirconia (ZrO.sub.2), silica (SiO.sub.2) or silicon nitride (Si.sub.3N.sub.4).
 10. The apparatus of claim 1, wherein the coil comprises a cooling pipe through which cooling water flows inside the coil.
 11. The apparatus of claim 1, wherein a thickness of the heat insulation member corresponds to twice the thickness of the susceptor.
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